

	GCE A/AS LEVEL – October/November 2007	9702	02
4	(a) brittle (b) Young modulus = stress / strain $= (9.5 \times 10^8) / 0.013$ $= 7.3 \times 10^{10} \text{ Pa}$ (allow $\pm 0.1 \times 10^{10} \text{ Pa}$) (c) stress = force / area (minimum) area $= (1.9 \times 10^3) / (9.5 \times 10^8)$ $= 2.0 \times 10^{-6} \text{ m}^2$ (max) area of cross-section $= (3.2 - 2.0) \times 10^{-6}$ $= 1.2 \times 10^{-6} \text{ m}^2$ (d) when bent, 'top' and 'bottom' edges have different extensions with thick rod, difference is greater (than with a thin rod) so breaks with less bending	B1 [1] C1 A1 [2] C1 C1 A1 [3] M1 A1 A0 [2]	
5	(a) amplitude between 6.5 squares and 7.5 squares on 3 peaks		B2